CHAPTER EXERCISE: Answers for this chapter start on page 293.

A calculator should NOT be used on the following questions.

1

$$30\left(x^3+\frac{1}{6}x^2+\frac{2}{3}x\right)=ax^3+bx^2+cx$$

In the equation above, a, b, and c are constants. If the equation is true for all values of x, what is the value of $a+b+c$?

2

$$\frac{2}{3}ax+3=\frac{8}{3}x+9b$$

In the equation above, a and b are constants. If the equation has infinitely many solutions, what is the value of $\frac{a}{b}$?

A) $\frac{3}{4}$

B) $\frac{4}{3}$

C) 6

D) 12

3

$$ax-b=3(2x+1)$$

In the equation above, a and b are constants. If the equation has no solution, which of the following could be the values of a and b ?

A) $a=2$ and $b=-3$

B) $a=2$ and $b=3$

C) $a=6$ and $b=-3$

D) $a=6$ and $b=3$

4

$$18x^2-8=2(ax+b)(ax-b)$$

In the equation above, a and b are constants. Which of the following could be the value of ab ?

A) 6

B) 9

C) 12

D) 36

5

$$x-\frac{1}{2}(3x+8)=2\left(2-\frac{1}{4}x\right)$$

How many solutions are there to the equation above?

A) The equation has no solutions.

B) The equation has infinitely many solutions.

C) The equation has exactly 1 solution.

D) The equation has exactly 2 solutions.

6

$$3x+a(3-2x)=12-7x$$

In the equation above, a is a constant. If the equation has no solutions, what is the value of a ?

A) -2

B) 2

C) 4

D) 5

7

If $(2x+3)(ax-5)=12x^2+bx-15$ for all values of x, what is the value of b ?

A) 6

B) 8

C) 10

D) 12

8

If $(x + 3y)^2 = x^2 + 9y^2 + 42$, what is the value of x^2y^2 ?

9

$$6x = x - 3x(2n - 1)$$

In the equation above, n is a constant. If the equation has infinitely many solutions, what is the value of n ?

A) $-\frac{2}{3}$

B) $-\frac{1}{3}$

C) $\frac{4}{3}$

D) $\frac{5}{3}$

10

If $\frac{ab + a}{b} = \frac{a}{b} + 5$ for all values of b, what is the value of a ?

11

If $\frac{1}{x} - \frac{1}{x - 4} = 1$, what is the value of x ?

12

If $n < 0$ and $4x^2 + mx + 9 = (2x + n)^2$, what is the value of $m + n$?

A) -15

B) -9

C) -3

D) 12

13

If $\frac{1}{x} + \frac{1}{y} = \frac{1}{p}$, what is x in terms of p and y ?

A) $p - y$

B) $\frac{py}{p + y}$

C) $\frac{py}{p - y}$

D) $\frac{py}{y - p}$

14

$$(x^3 + kx^2 - 3)(x - 2) = x^4 + 7x^3 - 18x^2 - 3x + 6$$

In the equation above, k is a constant. If the equation is true for all values of x, what is the value of k ?

A) -9

B) 5

C) 7

D) 9

15

$$\frac{5}{x+3} - \frac{2}{x-2} = \frac{ax-b}{(x+3)(x-2)}$$

The equation above is true for all $x > 2$, where a and b are constants. What is the value of $a + b$?

A) 7

B) 13

C) 19

D) 21

16

$$\frac{4}{x-1} + \frac{2}{x+1} = \frac{35}{x^2-1}$$

If $x > 1$, what is the solution to the equation above?

17

The equation $(2x - b)(7x + b) = 14x^2 - cx - 16$ is true for all values of x, where b and c are constants. If $b > 0$, what is the value of c ?

A) -20

B) 20

C) 28

D) 36

18

$$\frac{3}{n-1} + \frac{2n}{n+1} = 3$$

If $n > 0$, for what value of n is the equation above true?

Chapter 9: More Equation Solving Strategies

CHAPTER EXERCISE:

1. $\boxed{55}$ $30\left(x^3+\frac{1}{6}x^2+\frac{2}{3}x\right)=30x^3+5x^2+20x.$
   Therefore, $a=30$, $b=5$, and $c=20$.
   $a+b+c=55$.

2. $\boxed{D}$ For an equation to have infinitely many solutions, both sides must be equivalent.
   Comparing the terms on both sides, $\frac{2}{3}a=\frac{8}{3}$ and $3=9b$. Solving these equations, we get $a=4$ and $b=\frac{1}{3}$. Therefore, $\frac{a}{b}=\frac{4}{\frac{1}{3}}=12$.

3. $\boxed{D}$ For an equation to have no solutions, the coefficients of the x terms must be the same on either side but the constants must be different. If we expand the right side, we get
   $$ax-b=6x+3$$
   Therefore, $a=6$ and $b\neq -3$. Only answer choice D satisfies these conditions. Choice C would result in infinitely many solutions since both sides would be equivalent.

4. $\boxed{A}$ Remember, $p^2-q^2=(p+q)(p-q)$. We can apply this factorization here once we take out a 2:
   $18x^2-8=2(9x^2-4)=2(3x+2)(3x-2)$, which equals $2(ax+b)(ax-b)$. Comparing the coefficients, $a=3$ and $b=2$. Therefore, $ab=(3)(2)=6$. Note that this factorization method assumes that the constants are positive, but that's okay since all the answer choices are positive. It's possible that $a=-3$ and $b=2$ or $a=3$ and $b=-2$, for instance, but these are cases that you generally don't need to worry about for this factorization.
   If you didn't use this factorization and instead expanded the right side, you would've gotten
   $$18x^2-8=2a^2x^2-2b^2$$
   Comparing the coefficients on either side, $18=2a^2$ and $8=2b^2$. Solving these equations gives $a=\pm 3$ and $b=\pm 2$. As you can see, we haven't assumed that a and b are positive here. But of the answer choices, only 6 is a possible value of ab (when $a=3$ and $b=2$ or when $a=-3$ and $b=-2$).

5. $\boxed{A}$ Expand both sides of the equation.
   $$x-\frac{3}{2}x-4=4-\frac{1}{2}x$$
   $$-\frac{1}{2}x-4=4-\frac{1}{2}x$$
   Let's rearrange the right side so that the terms line up with the left side.
   $$-\frac{1}{2}x-4=-\frac{1}{2}x+4$$
   Now it's easy to see that the coefficients of the x terms are the same but the constants are different (-4 vs. 4). Therefore, the equation has no solutions.

6. $\boxed{D}$ For an equation to have no solutions, the coefficients of the x terms must be the same on either side but the constants must be different. First, let's expand the left side.
   $$3x+3a-2ax=12-7x$$
   $$3a-2ax=12-10x$$
   Comparing the coefficients of the x terms, $-2a=-10$, $a=5$. Note that a is NOT equal to 4 since the goal is not for the constants to be the same.

7. $\boxed{B}$ Expand the right side.
   $$(2x+3)(ax-5)=12x^2+bx-15$$
   $$2ax^2+3ax-10x-15=12x^2+bx-15$$
   $$2ax^2+(3a-10)x-15=12x^2+bx-15$$
   Comparing both sides, $2a=12$ and $b=3a-10$, which yields $a=6$ and $b=3a-10=3(6)-10=8$.

8. [49] Expand the left side:

$$x^2 + 6xy + 9y^2 = x^2 + 9y^2 + 42$$
$$\cancel{x^2} + 6xy + \cancel{9y^2} = \cancel{x^2} + \cancel{9y^2} + 42$$
$$6xy = 42$$
$$xy = 7$$
$$x^2y^2 = 49$$

9. [B] For an equation to have infinitely many solutions, both sides must be equivalent. First, let's expand the right side of the equation:

$$6x = x - 6nx + 3x$$
$$6x = 4x - 6nx$$
$$2x = -6nx$$

Now when we compare the coefficients on both sides, we get $2 = -6n$, which gives $n = \dfrac{2}{-6} = -\dfrac{1}{3}$.

10. [5] Multiply both sides by b.

$$ab + a = a + 5b$$
$$ab = 5b$$
$$a = 5$$

11. [2] Multiply both sides by $x(x-4)$.

$$(x-4) - x = x(x-4)$$
$$-4 = x^2 - 4x$$
$$0 = x^2 - 4x + 4$$
$$0 = (x-2)^2$$

We can see that $x = 2$.

12. [A] Expanding the right side,

$$4x^2 + mx + 9 = 4x^2 + 4nx + n^2$$

Comparing both sides, we see that

$$9 = n^2 \text{ and } m = 4n$$

Therefore, $n = -3$ and $m = 4(-3) = -12$

$$m + n = -12 + (-3) = -15$$

13. [D] Multiply both sides by xyp.

$$\frac{1}{x} + \frac{1}{y} = \frac{1}{p}$$
$$yp + xp = xy$$
$$yp = xy - xp$$
$$yp = x(y-p)$$
$$\frac{yp}{y-p} = x$$

14. [D] Expand the left side of the equation.

$$(x^3 + kx^2 - 3)(x-2)$$
$$= x^4 + kx^3 - 3x - 2x^3 - 2kx^2 + 6$$
$$= x^4 + (k-2)x^3 - 2kx^2 - 3x + 6$$

Comparing this to $x^4 + 7x^3 - 18x^2 - 3x + 6$, we can see that $k - 2 = 7$ and $-2k = -18$. In both cases, $k = 9$.

15. [C] Multiply both sides by $(x+3)(x-2)$. We get

$$5(x-2) - 2(x+3) = ax - b$$
$$5x - 10 - 2x - 6 = ax - b$$
$$3x - 16 = ax - b$$

Comparing the coefficients on either side, $a = 3$ and $b = 16$. Therefore, $a + b = 3 + 16 = 19$.

16. [$\frac{11}{2}$] Notice that $x^2 - 1 = (x+1)(x-1)$ on the right hand side. It's then easy to see that we should multiply both sides by $(x+1)(x-1)$.

$$4(x+1) + 2(x-1) = 35$$
$$4x + 4 + 2x - 2 = 35$$
$$6x + 2 = 35$$
$$6x = 33$$
$$x = \frac{11}{2}$$

17. [B] Expand the left hand side.

$$(2x - b)(7x + b) = 14x^2 - cx - 16$$
$$14x^2 + 2bx - 7bx - b^2 = 14x^2 - cx - 16$$
$$14x^2 - 5bx - b^2 = 14x^2 - cx - 16$$

Comparing both sides, we can see that $b = 4$ (b cannot be -4 because $b > 0$).
$c = 5b = 5(4) = 20$.

18. [3] Multiply both sides by $(n - 1)(n + 1)$.

$$3(n + 1) + 2n(n - 1) = 3(n + 1)(n - 1)$$
$$3n + 3 + 2n^2 - 2n = 3(n^2 - 1)$$
$$2n^2 + n + 3 = 3n^2 - 3$$
$$0 = n^2 - n - 6$$
$$0 = (n - 3)(n + 2)$$

$n = 3$ or -2. Because $n > 0$, $n = 3$.